Tarun Kumar E

Portfolio | GitHub | LinkedIn

Email: tarunkk.ds@gmail.com
Mobile: +91 9840373794

Professional Summary

BCA (Data Science) graduate with hands-on experience in Python, Large Language Models (LLMs), AI agents, and Generative AI workflows. Proven foundation in API integration, Retrieval-Augmented Generation (RAG) concepts, prompt engineering, and intelligent automation using tools like n8n. Equipped with strong analytical, logical reasoning, and problem-solving skills to build practical AI solutions.

Education

SRM Institute of Science and Technology — Chennai, Tamil Nadu
Bachelor of Computer Application (Data Science), GPA: 9.35 — *June 2023 – May 2026*
Relevant Coursework: *Data Science, Data Wrangling, DBMS, Natural Language Processing, Machine Learning, Deep Learning*

Prasan Vidhya Bala Mandir Hr. Sec. School — Chengalpattu, Tamil Nadu
Higher Secondary Education (12th), GPA: 8.95 — *2022 – 2023*

Skills Summary

- **Languages**: SQL, Python, HTML
- **Frameworks & Libraries**: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, NLTK, Flask, REST API, Matplotlib
- **AI & Generative AI**: AI Agents, Agent Workflows, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP) concepts, LLMs, Prompt Engineering, LangGraph, Ollama
- **Data & Vector Search**: ChromaDB, Data Profiling, ETL, REST APIs, API Integration
- **Tools & Platforms**: Power BI, Git, CI/CD, AWS (S3, IAM, EC2), Jupyter Notebook, VS Code, Windows
- **Soft Skills**: Analytical Thinking, Problem Solving, Communication, Stakeholder Collaboration, Leadership, Time Management

Internship

Rubixe Innovative Tech Solutions — Bengaluru
Data Engineer Intern — *Oct - Dec 2025*

- Built reusable Python-based data pipelines to automate data preprocessing, transformation, feature engineering, and downstream ML workflows.
- Performed data cleaning, preprocessing, feature selection, and transformation to prepare structured datasets for analytical and predictive workflows.
- Developed modular Scikit-learn pipelines integrating preprocessing, feature engineering, model training, and prediction steps into repeatable workflows.
- Automated model evaluation workflows and compared multiple algorithms using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix to identify optimal models.

Projects

Advanced RAG Documentation Assistant
Python, LangGraph, ChromaDB, Sentence Transformers, BM25

- Developed a Retrieval-Augmented Generation (RAG) assistant to answer documentation queries with context-grounded responses and source citations.
- Implemented hybrid search using BM25 keyword retrieval and ChromaDB vector search, combining results through Reciprocal Rank Fusion (RRF).
- Built a stateful LangGraph pipeline for document retrieval, prompt construction, and LLM-based answer generation using GPT-4o-mini.

Healthcare MCP Server – AI Powered
Python, FastMCP, SQLite, SQL, Streamable HTTP

- Built an MCP server exposing 4 reusable tools for patient risk scoring, readmission analysis, database schema inspection, and SQL querying.
- Implemented rule-based clinical risk classification using diagnoses, medications, lab results, and visit history to identify potential high-risk patients.
- Enabled AI clients to access SQLite healthcare data through Streamable HTTP, with validated inputs and controlled, read-only SQL execution.

Certifications

- **AWS Generative AI Foundation**: Completed AWS Generative AI Foundation course covering LLM fundamentals, prompt engineering, and applied AI workflows on AWS.
- **Mastering Generative AI (GeeksforGeeks)**: Completed hands-on training in LLM fundamentals, prompt engineering, and practical generative AI applications.
- **Google Data Analytics Certificate**: Trained in data cleaning, analysis, visualization, and SQL to derive actionable business insights.
- **Research Methodology in Data Science**: Developed an understanding of data collection, statistical validation, and experimental evaluation techniques used in data science. Gained exposure to structured research workflows and best practices for AI/ML projects.

Honors and Awards

- **Academic Excellence Award**: Secured First Rank in Mathematics at college level.
- **Merit Scholarship**: Received NGO-sponsored merit scholarship for academic excellence.
- **Club Excellence Award**: Recognized for outstanding performance and significant contributions to club growth and engagement.

Volunteer Experience

- **Tamil Mandram Club**: Assisted in planning and executing college events while coordinating logistics and participant engagement for smooth operations.